

Consistent Objectives for Self-Improving Systems

Absolute, Relative and Evolutionary Assessment Over Short and Long Horizons

Peter Cotton*

Abstract

A self-improving system accepts or rejects changes to itself by a per-step objective, and is judged by where it ends up. We ask which per-step objectives select the same trajectories as that long-term goal, comparing absolute scores, win rates against a predecessor or a fixed anchor, selection of a population’s best, and relative wealth under compounding. Improvement in a fixed absolute score, less a constant cost per step, is the only per-step reward consistent for every feasible set of trajectories. Compounded relative wealth is consistent with expected log return, and the other relative schemes reward churn, distort attitudes to risk, reward variance, or record evaluator drift as progress.

1 Introduction

Each proposed change to a self-improving system, whether to a prompt, a memory, a piece of code or a set of weights, is accepted or rejected on evidence available now. The system as a whole is judged later, by what it has become. The two judgements coincide only for particular choices of the per-step objective.

Some objectives in use are *absolute*, such as a benchmark score, a test pass rate, or a proper scoring rule against resolved outcomes. Others are *relative*. Self-play and self-rewarding language models score a version against its predecessor (Yuan et al., 2024). Leaderboards rank versions by pairwise preference (Chiang et al., 2024). Evolutionary search over agents keeps the survivors of a population (Zhang et al., 2025; Iacob et al., 2026).

The literature on each line is large. Reward shaping characterises which per-step rewards preserve optimal policies (Ng et al., 1999). Goodhart’s law, reward hacking and reward-model overoptimisation describe what goes wrong when a proxy is optimised hard (Manheim and Garrabrant, 2018; Skalse et al., 2022; Gao et al., 2022; Karwowski et al., 2023). Non-transitive games show that beating a predecessor need not mean getting stronger (Balduzzi et al., 2019; Czarnecki et al., 2020). Forecasting competitions break the propriety of scoring rules (Lichtendahl and Winkler, 2007; Witkowski et al., 2021). Evolutionary finance shows which beliefs survive market selection (Blume and Easley, 1992; Sandroni, 2000; Blume and Easley, 2006).

The literature rarely puts these results side by side. Recent frameworks classify self-improving systems by whether their standard is grounded outside the agent (Tang et al., 2026) and catalogue evaluator drift as a safety risk (Gong et al., 2026). The Red Queen Gödel Machine holds the evaluator fixed within epochs so that improvement guarantees hold per epoch (Iacob et al., 2026). None of them compares the objectives themselves for consistency between the step and the horizon. We make that comparison with elementary propositions, each checked numerically.

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2 Setup

A system passes through versions $\theta_0, \theta_1, \dots, \theta_T$. At each step it chooses θ_{t+1} from a feasible set $N(\theta_t)$, which may or may not include staying put. The *long-term objective* is a utility of the final version, $U(\theta_T)$. For a discounted sum of deployed utility, the potential term in Proposition 1 becomes $\gamma\Phi(\theta') - \Phi(\theta)$ (Ng et al., 1999). A *per-step reward* $R(\theta, \theta')$ scores a single change, and the system maximises $\sum_{t < T} R(\theta_t, \theta_{t+1})$, or a greedy approximation to it.

Definition 1 (Horizon consistency). A per-step reward R is *horizon consistent* with U if, for every horizon T , every starting version and every feasibility structure N , the trajectories that maximise $\sum_{t < T} R(\theta_t, \theta_{t+1})$ are exactly those that maximise $U(\theta_T)$.

The definition asks for agreement on the whole set of maximisers. A reward that agrees with U on most problems but not all fails on the problems a thorough search finds, which is the mechanism behind Goodhart’s law (Manheim and Garrabrant, 2018).

3 Absolute assessment

Proposition 1 (Only potential differences are consistent). A *per-step reward* R is *horizon consistent* with U if and only if

$$R(\theta, \theta') = \Phi(\theta') - \Phi(\theta) + k$$

for some constant k and some strictly increasing transformation Φ of U .

The reward for a change is the rise in a monotone transformation of the long-term score, plus the same constant for every step.

Proof. Sufficiency is telescoping: the sum over a path of length T equals $\Phi(\theta_T) - \Phi(\theta_0) + kT$, which ranks endpoints as U does.

For necessity, take $T = 1$ and let $N(\theta_0)$ be an arbitrary set of versions: $R(\theta_0, \cdot)$ must rank versions as U does. Take $T = 2$ with $N(a) = \{b, b'\}$ and $N(b) = N(b') = \{c\}$. The two feasible paths $a \rightarrow b \rightarrow c$ and $a \rightarrow b' \rightarrow c$ share their endpoint, so both maximise U and they must tie, and $R(a, b) + R(b, c)$ cannot depend on b .

Write it as $h(a, c)$. Fixing $c = c_0$ gives $R(a, b) = \alpha(a) - \beta(b)$ with $\alpha(a) = h(a, c_0)$ and $\beta(b) = R(b, c_0)$. Substituting back, $\alpha(b) - \beta(b)$ must be a constant k , so $R(a, b) = \alpha(a) - \alpha(b) + k$. Set $\Phi = -\alpha$. The $T = 1$ condition makes Φ increasing in U . $\square$

A negative k is a cost per step, and $-k$ can be read as the price of a computation in the sense of Russell and Wefald (1991). Proposition 1 is the undiscounted case of the shaping theorem of Ng et al. (1999), specialised to a system whose state is its own version. Under uncertainty only affine Φ preserve rankings of expected utility, so in practice the consistent reward is improvement in the long-term score itself, less a cost.

The score must be proper. If U is an expected score of probabilistic forecasts against resolved outcomes, a strictly proper scoring rule makes truthful reporting optimal (Brier, 1950; Gneiting and Raftery, 2007). An improper score adds a second inconsistency, between the forecasts the system holds and the forecasts it reports. Training on resolved outcomes with a proper score (Turtel et al., 2026) meets both conditions.

Greedy acceptance is a separate failure. Proposition 1 concerns the objective, not the search. A system that accepts the change with the largest immediate improvement can still be arbitrarily far from optimal. If one change gains ϵ and leads nowhere while another gains nothing and unlocks a gain of one, greedy acceptance takes the first. The remedy is to value a change by its effect on what can follow, which is the value of computation (Russell and Wefald, 1991; Hay et al., 2012). For changes that run in parallel, the corresponding rule keeps a path while its future pivotal value exceeds its carrying cost (Cotton, 2026a).

4 Relative assessment against a predecessor

Pairwise comparison is attractive because a judge can often say which of two outputs is better when it cannot score either. Suppose comparisons follow a Bradley–Terry model (Bradley and Terry, 1952), so that θ' beats θ with probability $\sigma(u(\theta') - u(\theta))$ for a latent strength u and the logistic function σ .

Proposition 2 (Win rate against the predecessor rewards churn). *Let $R(\theta, \theta') = \sigma(u(\theta') - u(\theta))$.*

1. *Accepting a change exactly when $R > \frac{1}{2}$ accepts exactly the changes that raise u .*
2. *Splitting any gain $D > 0$ into two steps of $D/2$ raises the cumulative reward from $\sigma(D) < 1$ to $2\sigma(D/2) > 1$. Two changes that leave u unchanged earn 1, more than any single change of any size.*
3. *R is not horizon consistent. There are feasibility structures in which the cumulative reward selects an endpoint of lower u .*

Proof. The first part is monotonicity of σ . The second uses $\sigma(0) = \frac{1}{2}$ and $\sigma < 1$. For the third, let the start have $u = 0$ and offer two routes over four steps: a jump to $u = 5$ followed by staying put, or a ladder of four unit steps to $u = 4$. The jump route earns $\sigma(5) + \frac{3}{2} \approx 2.49$ and the ladder $4\sigma(1) \approx 2.92$. The cumulative win rate takes the ladder to the worse endpoint. $\square$

The ordinal acceptance rule is sound and the cumulative objective is not. A system trained to maximise accumulated wins against its own predecessors is paid for the number of changes, not for their size. Without a transitive latent strength the failure is worse: in a game with strategic cycles, every step can beat its predecessor while the sequence returns to its start (Balduzzi et al., 2019). Real games combine a transitive axis with many cycles at each level of strength (Czarnecki et al., 2020), which is why population methods evaluate against mixtures rather than the last version (Lanctot et al., 2017; Balduzzi et al., 2018).

5 Relative assessment against a fixed anchor

Comparing every version with one fixed reference, $R = \sigma(u(\theta) - u_{\text{ref}})$, is a strictly increasing transformation of u . It is horizon consistent when outcomes are certain. It is not affine, so it changes the system’s attitude to risk.

Proposition 3 (A fixed anchor makes laggards risk-seeking). *If the uncertain strength of a proposed version has support entirely below u_{ref} , any mean-preserving spread raises the expected win rate against the anchor. If the support lies entirely above, any mean-preserving spread lowers it.*

Proof. The logistic function σ is convex on $(-\infty, 0)$ and concave on $(0, \infty)$, and Jensen’s inequality gives both claims. $\square$

In a normal example with the anchor three units ahead, nearly all of the mass lies where σ is convex. Widening the standard deviation of a proposal from 0.1 to 1 raises its expected win rate from 0.048 to 0.069. Three units behind, the same spread lowers it from 0.952 to 0.931. A system that trails its anchor is paid to gamble, and a system that leads is paid to stop experimenting.

6 Relative assessment within a population

Evolutionary and tournament methods keep the best of several candidates, judged on realised scores. The candidate with the highest expected score is not the one most likely to post the highest realised score.

Proposition 4 (Selection of the realised best rewards variance). *Let candidate i post $S_i = \mu_i + s_i Z_i$ with independent standard normal Z_i . For two candidates,*

$$P(S_1 > S_2) = \mathcal{N}\left(\frac{\mu_1 - \mu_2}{\sqrt{s_1^2 + s_2^2}}\right),$$

where $\mathcal{N}$ is the standard normal distribution function. The probability increases in s_1 whenever $\mu_1 < \mu_2$.

Proof. The difference $S_1 - S_2$ is normal with mean $\mu_1 - \mu_2$ and variance $s_1^2 + s_2^2$, which gives the formula. When $\mu_1 < \mu_2$ the argument of $\mathcal{N}$ is negative, and raising s_1 moves it towards zero. $\square$

A trailing candidate gains by being noisier. With $\mu_2 - \mu_1 = 0.5$ and $s_2 = 1$, raising s_1 from 0.5 to 4 lifts its chance of selection from 0.33 to 0.45. Take a field of ten candidates with unit standard deviation and equal means, except that one is better by half a unit. Doubling the standard deviation of another candidate makes it the most likely winner: it is selected 23% of the time, against 17% for the better candidate. Closed forms and Monte Carlo agree to within 2×10^{-3} in win probability.

Forecasters competing for a single prize report more extreme probabilities than they believe (Lichtendahl and Winkler, 2007), for the same reason, and incentive-compatible competitions are designed to remove the distortion (Witkowski et al., 2021). The selected alternative’s realised score overstates its value (Smith and Winkler, 2006). In backtests the best of many strategies needs a deflated threshold (Bailey and López de Prado, 2014), and LLM-discovered trading strategies fail once the agent’s own trial count is accounted for (Gençay, 2026).

When rewarding variance is right. If the long-term objective is itself an expected maximum, selection by realised maximum is consistent with it. Best-of- k deployment and pass@ k pay the best of several attempts, and there variance has value (Cotton, 2026a,b). Selection by maximum suits a deployment that pays the maximum, and misleads one that pays the mean.

Finite populations and spite. Relative fitness in a finite population rewards lowering a rival’s payoff as much as raising one’s own. [Hamilton \(1970\)](#) identified spiteful behaviour, and [Schaffer \(1988\)](#) showed that evolutionarily stable strategies in finite populations maximise relative rather than absolute payoff. A population of agents judged against one another can therefore select changes that make every member worse in absolute terms.

7 Relative assessment under compounding

Relative assessment becomes consistent with an absolute objective when wealth compounds. Let strategies compound wealth by gross returns $R_{i,t}$, independent over time with finite $\mathbb{E}|\log R_i|$, and let assessment be relative: the share $w_{i,t}/\sum_j w_{j,t}$.

Proposition 5 (Relative wealth selects the largest expected log return). *If $\mathbb{E} \log R_1 > \mathbb{E} \log R_j$ for all $j \neq 1$, the share of strategy 1 tends to one almost surely, whatever the expected returns $\mathbb{E} R_j$.*

Proof. By the strong law of large numbers, $t^{-1} \log w_{i,t} \rightarrow \mathbb{E} \log R_i$ almost surely. The ratio $w_{j,t}/w_{1,t}$ therefore tends to zero for every $j \neq 1$. $\square$

The proposition is the selection side of [Kelly \(1956\)](#) and [Breiman \(1961\)](#). It separates the short-term and long-term readings of an absolute objective.

A strategy with returns 1.6 or 0.5 with equal probability has $\mathbb{E} R = 1.05$ and $\mathbb{E} \log R = -0.11$. A strategy with returns 1.1 or 0.95 has $\mathbb{E} R = 1.025$ and $\mathbb{E} \log R = +0.022$. The first wins on one-step expected return, and its expected wealth after 400 steps exceeds 10^8 . It loses the population in every one of 20,000 simulated runs, since its median log wealth falls by 0.11 a step. Expected wealth and typical wealth diverge because wealth is not ergodic ([Peters, 2019](#)).

Beliefs. In markets the same selection operates on forecasts. Agents survive in proportion to how well their beliefs track the truth ([Blume and Easley, 1992](#); [Sandroni, 2000](#)), with qualifications for incomplete markets ([Blume and Easley, 2006](#)). A forecaster paid by the log score relative to a market grows its share at the rate of its log-score advantage, so compounded relative assessment recovers the log score, a strictly proper absolute objective. [Cotton \(2022, ch. 5\)](#) considers agents that solicit, compensate and combine the predictions of other agents. When such compensation is relative and compounds, it belongs to this class.

Absolute selection. When fitness does not depend on the population, selection raises mean fitness. Under replicator dynamics ([Taylor and Jonker, 1978](#)) with constant fitnesses, mean fitness rises at a rate equal to the variance of fitness. This is the simplest case of Fisher’s fundamental theorem, whose precise statement is due to [Price \(1972\)](#). The guarantee does not survive frequency-dependent fitness, the setting of evolutionary game theory ([Maynard Smith and Price, 1973](#)).

8 Moving evaluators

A system that helps write its own evaluator breaks the telescoping in [Proposition 1](#). Let Φ_t be the evaluator in force at step t , and let the system record the improvement $\Phi_t(\theta_{t+1}) - \Phi_t(\theta_t)$ at each step.

Proposition 6 (Measured progress includes evaluator drift).

$$\sum_{t < T} [\Phi_t(\theta_{t+1}) - \Phi_t(\theta_t)] = \Phi_T(\theta_T) - \Phi_0(\theta_0) - \sum_{t < T} [\Phi_{t+1}(\theta_{t+1}) - \Phi_t(\theta_{t+1})].$$

The identity follows by adding and subtracting $\Phi_{t+1}(\theta_{t+1})$ at each step. Recorded progress equals the change in the current evaluator’s opinion, less the drift in the evaluator at the versions actually visited. With a fixed anchored evaluator the drift term vanishes and recorded progress is real.

The drift term has either sign. An evaluator that tightens as the system improves makes recorded progress understate the change, which is the intent of co-evolving evaluators (Iacob et al., 2026). An evaluator that comes to share the system’s errors, as when a model judges its own outputs (Yuan et al., 2024; Huang et al., 2023) or trains on them (Shumailov et al., 2024), can make the anchored score stand still while the measured one rises. Holding the evaluator fixed within an epoch restores telescoping within the epoch, and accumulating real data rather than replacing it avoids collapse in the settings studied by Gerstgrasser et al. (2024).

9 Objectives compared

Per-step objective	Consistent with the long-term goal	Failure when optimised
Improvement in a fixed proper score, less cost	Yes, for every feasibility structure	Greedy search misses option value
Win rate against the predecessor	Ordinal acceptance only	Churn, and cycles if non-transitive
Win rate against a fixed anchor	Yes under certainty	Risk-seeking behind, risk-averse ahead
Best of a population on realised score	Only for expected-maximum goals	Rewards variance, winner’s curse
Relative fitness in a finite population	No	Spite
Relative wealth under compounding	Yes, for expected log return	One-step expected value loses
Improvement under a moving evaluator	Only with a fixed anchor	Drift recorded as progress

10 Closing

Neither absolute nor relative assessment dominates. Absolute improvement in a fixed proper score is the one per-step objective consistent for every feasibility structure, but it needs a score that can be computed, and a search that values option value as well as immediate gain. Relative assessment is cheaper and often the only signal available. It is consistent in two cases, ordinal acceptance under a transitive latent strength and multiplicative compounding. Elsewhere it produces churn, distorted risk, rewarded variance, spite or drift.

The trade-off is not particular to machine learning. Rank-order tournaments for workers (Lazear and Rosen, 1981), forecasting contests and natural selection face the same choice between

paying for level and paying for rank. We leave to future work the empirical measurement of these effects in a running self-improving system, where the drift term in Proposition 6 can be estimated by re-scoring past versions under the current evaluator.

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